CORE PURE / DIFFERENTIAL EQUATIONS

Systems of Differential Equations

Mark scheme 12 questions · 171 marks

Questions

Question	Marks by part	Marks	Page
Question 1	(a) 3 (b) 7	10	4
Question 2		11	5
Question 3		11	6
Question 4	(a) 2 (b) 11	13	7
Question 5		10	8
Question 6		11	9
Question 7	(a) 3 (b) 6 (c) 2 (d) 6 (e) 1	18	10
Question 8	(a) 3 (b) 4 (c) 3 (d) 7	17	12
Question 9	(a) 2 (b) 3 (c) 11	16	14
Question 10	(a) 1 (b) 3 (c)(i) 2 (c)(ii) 1 (c)(iii) 3 (d) 5 (e) 1	16	16
Question 11	(a) 3 (b) 7 (c) 2 (d) 2 (e) 1	15	18
Question 12	(a)(i) 5 (a)(ii) 2 (b)(i) 3 (b)(ii) 1 (c)(i) 2 (c)(ii) 4 (c)(iii) 3 (d)(i) 1 (d)(ii) 2	23	20
	Total	**171**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}y}{\mathrm{d}t} = 5\frac{\mathrm{d}x}{\mathrm{d}t}$ $\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6x - 2y = 5\frac{\mathrm{d}x}{\mathrm{d}t}$	**M1**	Differentiates the first equation and replaces $\frac{\mathrm{d}y}{\mathrm{d}t}$ using the second. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6x - 2\left(5x - \frac{\mathrm{d}x}{\mathrm{d}t}\right) = 5\frac{\mathrm{d}x}{\mathrm{d}t}$	**dM1**	Substitutes y from the first equation to reach an equation in x and its derivatives only. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} - 4x = 0$	**A1 AG**	Correct substituted equation leading without error to the printed result; collected steps accepted.
		[3]	
1(b)	$\lambda^2 - 3\lambda - 4 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any method, calculator allowed.
	$\lambda = 4,\ -1$ $x = Ae^{4t} + Be^{-t}$	**A1**	
	$\frac{\mathrm{d}x}{\mathrm{d}t} = 4Ae^{4t} - Be^{-t}$ $y = 5\left(Ae^{4t} + Be^{-t}\right) - \left(4Ae^{4t} - Be^{-t}\right)$	**M1**	Differentiates their x, of the form $Ae^{pt} + Be^{qt}$, and substitutes into $y = 5x - \frac{\mathrm{d}x}{\mathrm{d}t}$. Condone sign slips.
	$y = Ae^{4t} + 6Be^{-t}$	**A1**	
	$A + B = 1, \quad A + 6B = 3$	**M1**	Substitutes $t = 0$, $x = 1$ and $y = 3$ into their x and y to form two equations in A and B.
	$\Rightarrow A = \ldots,\ B = \ldots$	**dM1**	Solves their two equations by any means, calculator allowed.
	$x = \frac{3}{5}e^{4t} + \frac{2}{5}e^{-t}$ $y = \frac{3}{5}e^{4t} + \frac{12}{5}e^{-t}$	**A1**	Both, oe.
		[7]	
	Total for Question 1	**10**	

Question	Scheme	Marks	Notes
2	$y = -\frac{\mathrm{d}x}{\mathrm{d}t} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ $\Rightarrow -\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - x = 2e^t$	**M1**	Differentiates one equation and substitutes from the other to reach a second-order equation in x alone, or y alone. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x = -2e^t$	**A1**	oe, or $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + y = 2e^t$.
	$t = 0 : \frac{\mathrm{d}x}{\mathrm{d}t} + 1 = 0 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -1$	**B1**	Or $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ at $t = 0$ where x is eliminated.
	$\lambda^2 + 1 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation of their second-order equation and solves it by any method.
	$\lambda = \pm i$ $x = A\cos t + B\sin t$	**A1**	oe, in the variable of their equation.
	$x = ae^t \Rightarrow ae^t + ae^t = -2e^t \Rightarrow a = \ldots$	**M1**	Substitutes a particular integral ae^t into their second-order equation and finds a. Condone sign slips.
	$x = A\cos t + B\sin t - e^t$	**A1**	
	$A - 1 = 0$ $\frac{\mathrm{d}x}{\mathrm{d}t} = -A\sin t + B\cos t - e^t$ $\Rightarrow B - 1 = -1 \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Uses both initial conditions in their general solution of the form $A\cos t + B\sin t +$ their particular integral, and its attempted derivative, to find A and B.
	$x = \cos t - e^t$	**A1**	
	$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t - e^t \Rightarrow y = -\frac{\mathrm{d}x}{\mathrm{d}t} = \ldots$	**M1**	Differentiates their x and uses $y = -\frac{\mathrm{d}x}{\mathrm{d}t}$, or any valid route to y leaving no undetermined constants.
	$y = \sin t + e^t$	**A1**	
	Total for Question 2	**11**	

Question	Scheme	Marks	Notes
3	$x = -\frac{\mathrm{d}y}{\mathrm{d}t} - y - 1 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t}$ $-\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t}$ $= 3\left(-\frac{\mathrm{d}y}{\mathrm{d}t} - y - 1\right) + 4y + 7$	**M1**	Makes x the subject of the second equation, differentiates and substitutes both into the first. Condone sign slips. Eliminating y instead, giving $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} + x = 3$, earns the corresponding marks.
	$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2\frac{\mathrm{d}y}{\mathrm{d}t} + y = -4$	**A1**	oe
	$\lambda^2 - 2\lambda + 1 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation of their second-order equation and solves it by any means.
	$\lambda = 1$ (repeated) $(y =)\ (A + Bt)e^t$	**A1**	
	$y = k \Rightarrow k = -4$	**B1**	
	$y = (A + Bt)e^t - 4$	**A1ft**	ft their complementary function plus their constant particular integral; must be $y = \ldots$.
	$y(0) = -3 \Rightarrow A - 4 = -3 \Rightarrow A = \ldots\ [= 1]$	**M1**	Substitutes $t = 0$, $y = -3$ into their general solution to find A.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = (A + Bt + B)e^t$ $x = -(A + Bt + B)e^t - (A + Bt)e^t + 4 - 1$	**M1**	Differentiates their y, with the product rule on $(A + Bt)e^t$, and substitutes into x made the subject of the second equation, or an equivalent relation from the system.
	$x = -(2A + 2Bt + B)e^t + 3$	**A1**	oe, in terms of A and B or with $A = 1$.
	$x(0) = 5 \Rightarrow -(2 + B) + 3 = 5 \Rightarrow B = \ldots$	**M1**	Substitutes $t = 0$, $x = 5$ into their x, or uses $\frac{\mathrm{d}y}{\mathrm{d}t} = -3$ at $t = 0$, to find B.
	$x = (2 + 8t)e^t + 3$ $y = (1 - 4t)e^t - 4$	**A1**	Both, oe.
	Total for Question 3	**11**	

Question	Scheme	Marks	Notes
4(a)	$x+2y-7=0,\ 2x+y-8=0 \Rightarrow x=\ldots,\ y=\ldots$	**M1**	Sets both expressions equal to zero and solves the simultaneous equations by any method.
	$(3,\ 2)$	**A1**	
		[2]	
4(b)	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \frac{\mathrm{d}x}{\mathrm{d}t} + 2\frac{\mathrm{d}y}{\mathrm{d}t}$ $= \frac{\mathrm{d}x}{\mathrm{d}t} + 4x + 2y - 16$ with $2y = \frac{\mathrm{d}x}{\mathrm{d}t} - x + 7$	**M1**	Differentiates one equation, substitutes the other and eliminates one variable to reach a second-order equation in a single variable. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} - 3x = -9$	**A1**	oe
	$\lambda^2 - 2\lambda - 3 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation of their second-order equation and solves it by factorising, formula or completing the square; calculator allowed.
	$Ae^{3t} + Be^{-t}$	**A1**	
	$-3k = -9 \Rightarrow k = 3$	**B1**	
	$x = Ae^{3t} + Be^{-t} + 3$	**A1ft**	ft their complementary function plus their constant particular integral.
	$A + B + 3 = 4$ $3A - B = 4 + 2(0) - 7$	**M1**	Forms two equations in A and B: $x(0) = 4$ and a second condition from the system at $t = 0$, e.g. $\frac{\mathrm{d}x}{\mathrm{d}t} = -3$ in their differentiated solution. Condone slips.
	$\Rightarrow A = \ldots,\ B = \ldots$	**dM1**	Solves their two equations for A and B; calculator allowed.
	$x = 3 - \frac{1}{2}e^{3t} + \frac{3}{2}e^{-t}$	**A1**	oe
	$2y =$ "$-\frac{3}{2}e^{3t} - \frac{3}{2}e^{-t}$" $-$ "$3 - \frac{1}{2}e^{3t} + \frac{3}{2}e^{-t}$" $+ 7$	**M1**	Substitutes their x and their $\frac{\mathrm{d}x}{\mathrm{d}t}$ into the first equation to find y, with their x of the form $pe^{\alpha t} + qe^{\beta t} + c$, numerical constants.
	$y = 2 - \frac{1}{2}e^{3t} - \frac{3}{2}e^{-t}$	**A1**	oe
		[11]	
	Total for Question 4	**13**	

Question	Scheme	Marks	Notes
5	$y = 2x - \frac{\mathrm{d}x}{\mathrm{d}t} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = 2\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ $2\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = x + 2\left(2x - \frac{\mathrm{d}x}{\mathrm{d}t}\right)$	**M1**	Makes one variable the subject of one equation, differentiates it and substitutes into the other, eliminating that variable. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0$	**A1**	oe, including the same equation in y.
	$\lambda^2 - 4\lambda + 5 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation of their second-order equation and solves it by any means, calculator allowed.
	$[\lambda = 2 \pm i]$ $x = e^{2t}(A\cos t + B\sin t)$	**A1**	Any letters for the constants; must be in terms of t.
	$t = 0 : \frac{\mathrm{d}x}{\mathrm{d}t} = 2(1) - 2 = 0$	**B1**	
	$\frac{\mathrm{d}x}{\mathrm{d}t} = 2e^{2t}(A\cos t + B\sin t)$ $\quad + e^{2t}(-A\sin t + B\cos t)$	**M1**	Differentiates their complementary function by the product rule, with both terms present. Condone sign slips in differentiating $\sin t$ and $\cos t$.
	$A = 1,\ 2A + B = 0 \Rightarrow B = \ldots$	**dM1**	Substitutes $t = 0$ into their x and their derivative, using $x = 1$ and their value of $\frac{\mathrm{d}x}{\mathrm{d}t}$, and solves for both constants.
	$f(t) = e^{2t}(\cos t - 2\sin t)$	**A1**	oe simplified, such as $e^{2t}\cos t - 2e^{2t}\sin t$.
	$y = 2e^{2t}(\cos t - 2\sin t) + 5e^{2t}\sin t$	**M1**	Finds the second function from their first using one of the given equations, or by solving its own second-order equation with both its initial conditions.
	$g(t) = e^{2t}(2\cos t + \sin t)$	**A1**	oe simplified, such as $2e^{2t}\cos t + e^{2t}\sin t$.
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6	$y = -\frac{\mathrm{d}x}{\mathrm{d}t} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ $\Rightarrow -\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - x = 2$	**M1**	Makes one variable the subject of one equation, differentiates and substitutes into the other, reaching a second-order equation in one variable. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x = -2$	**A1**	oe
	$\lambda^2 + 1 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation of their second-order equation and solves it by any means.
	$\lambda = \pm i$ $(x =)\ A\cos t + B\sin t$	**A1**	
	$x = k \Rightarrow k = -2$	**B1**	
	$x = A\cos t + B\sin t - 2$	**A1ft**	ft their complementary function plus their constant particular integral, as $x = \ldots$ with two arbitrary constants.
	$t = \frac{\pi}{2}$: $\frac{\mathrm{d}x}{\mathrm{d}t} = -y = -2$	**B1**	soi, e.g. by $-A = -2$.
	$1 = A\cos\frac{\pi}{2} + B\sin\frac{\pi}{2} - 2$ "-2" $= -A\sin\frac{\pi}{2} + B\cos\frac{\pi}{2}$ $\Rightarrow A = \ldots,\ B = \ldots$	**M1**	Substitutes $t = \frac{\pi}{2}$, $x = 1$ into their general solution and their derivative value into its derivative, finding both constants. Condone differentiation sign slips.
	$x = 2\cos t + 3\sin t - 2$	**A1**	
	$y = -\frac{\mathrm{d}x}{\mathrm{d}t} = -(-2\sin t + 3\cos t)$	**M1**	Differentiates their x and uses $y = -\frac{\mathrm{d}x}{\mathrm{d}t}$, or another relation from the system, without new arbitrary constants. Condone sign slips.
	$y = 2\sin t - 3\cos t$	**A1**	
	Total for Question 6	**11**	

Question	Scheme	Marks	Notes
7(a)	$\frac{\mathrm{d}^2w}{\mathrm{d}t^2} = 4\frac{\mathrm{d}w}{\mathrm{d}t} - \frac{\mathrm{d}s}{\mathrm{d}t}$ $= 4\frac{\mathrm{d}w}{\mathrm{d}t} - (6w - s - 48e^{-t})$	**M1**	Differentiates the first equation and uses the second to replace $\frac{\mathrm{d}s}{\mathrm{d}t}$, or equivalent. Condone sign slips.
	$= 4\frac{\mathrm{d}w}{\mathrm{d}t} - 6w$ $+ \left(4w - \frac{\mathrm{d}w}{\mathrm{d}t}\right) + 48e^{-t}$	**dM1**	Eliminates s using $s = 4w - \frac{\mathrm{d}w}{\mathrm{d}t}$, reaching an equation in w and its derivatives only.
	$\frac{\mathrm{d}^2w}{\mathrm{d}t^2} - 3\frac{\mathrm{d}w}{\mathrm{d}t} + 2w = 48e^{-t}$	**A1 AG**	Correct substituted equation leading without error to the printed result; collected steps accepted.
		[3]	
7(b)	$\lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by factorising with product of constants 2, the formula or completing the square; calculator accepted.
	$\lambda = 1,\ 2$ $(w =)\ Ae^{t} + Be^{2t}$	**A1**	
	$w = ae^{-t}$	**B1**	
	$ae^{-t} + 3ae^{-t} + 2ae^{-t} = 48e^{-t} \Rightarrow a = \ldots$	**M1**	Differentiates their trial form twice, substitutes into the equation and equates coefficients to find a. Condone sign slips.
	$8e^{-t}$	**A1**	
	$w = Ae^{t} + Be^{2t} + 8e^{-t}$	**A1ft**	ft their complementary function of the form $Ae^{\alpha t} + Be^{\beta t}$ plus their particular integral. Must be $w = \ldots$ in terms of t.
		[6]	
7(c)	$\frac{\mathrm{d}w}{\mathrm{d}t} = Ae^{t} + 2Be^{2t} - 8e^{-t}$ $s = 4(\ldots) - (\ldots)$	**M1**	Differentiates their w and substitutes into $s = 4w - \frac{\mathrm{d}w}{\mathrm{d}t}$. Condone coefficient slips.
	$s = 3Ae^{t} + 2Be^{2t} + 40e^{-t}$	**A1**	Constants must match those in w.
		[2]	
7(d)	$30 = A + B + 8,\ 110 = 3A + 2B + 40$	**M1**	Substitutes $t = 0$, $w = 30$, $s = 110$ into their w and s, forming two equations in A and B.

Question 7 continued

Question	Scheme	Marks	Notes
	$A + B = 22,\ 3A + 2B = 70 \Rightarrow A = \ldots,\ B = \ldots$	**dM1**	Solves their two linear equations by any method; calculator accepted.
	$A = 26,\ B = -4$	**A1**	
	$26e^T - 4e^{2T} + 8e^{-T} = 0$ $\Rightarrow 2x^3 - 13x^2 - 4 = 0,\ x = e^T$	**M1**	Sets their $w = 0$ at $t = T$ and forms a cubic in e^T, or solves the equation in T directly. Need not be simplified.
	$x = 6.547\ldots$	**B1**	awrt 6.55; implied by a correct T.
	$T = \ln x = 1.879$	**A1**	
		[6]	
7(e)	The model predicts negative values of w after T	**E1**	States a limitation specific to this model, e.g. negative w or s predicted, s not an integer, unbounded growth. Not for "not accurate" alone.
		[1]	
	Total for Question 7	**18**	

Question	Scheme	Marks	Notes
8(a)	$\frac{\mathrm{d}^2s}{\mathrm{d}t^2} = -\frac{1}{20}\frac{\mathrm{d}m}{\mathrm{d}t} - \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t}$	**M1**	Differentiates the $\frac{\mathrm{d}s}{\mathrm{d}t}$ equation, or m made its subject, with respect to t. Condone coefficient slips.
	$= -\frac{1}{20}\left(\frac{1}{4}s\right) - \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t}$	**dM1**	Replaces $\frac{\mathrm{d}m}{\mathrm{d}t}$ by $\frac{1}{4}s$ to eliminate m.
	$\frac{\mathrm{d}^2s}{\mathrm{d}t^2} + \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t} + \frac{1}{80}s = 0$	**A1 AG**	Correct substituted equation leading without error to the printed result; combined rearrangement accepted.
		[3]	
8(b)	$\lambda^2 + \frac{1}{10}\lambda + \frac{1}{80} = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by formula with their values, completing the square or calculator.
	$\lambda = -\frac{1}{20} \pm \frac{1}{10}i$	**A1**	
	$s = e^{-t/20}\left(A\cos\frac{t}{10} + B\sin\frac{t}{10}\right)$	**dM1 A1**	**dM1** Uses $e^{pt}(A\cos qt + B\sin qt)$ for their roots $p \pm qi$; a complex exponential form does not score.
		[4]	
8(c)	$m = -20\left(\frac{\mathrm{d}s}{\mathrm{d}t} + \frac{1}{10}s\right)$ $\frac{\mathrm{d}s}{\mathrm{d}t} = -\frac{1}{20}e^{-t/20}\left(A\cos\frac{t}{10} + B\sin\frac{t}{10}\right)$ $+ e^{-t/20}\left(\frac{B}{10}\cos\frac{t}{10} - \frac{A}{10}\sin\frac{t}{10}\right)$	**M1**	Makes m the subject of the second equation and differentiates their s, of the form $e^{pt}(A\cos qt + B\sin qt)$, by the product rule. Condone sign and coefficient slips.
	$m = -20e^{-t/20}\Big[\left(\frac{A}{20} + \frac{B}{10}\right)\cos\frac{t}{10}$ $+ \left(\frac{B}{20} - \frac{A}{10}\right)\sin\frac{t}{10}\Big]$	**dM1**	Substitutes their s and $\frac{\mathrm{d}s}{\mathrm{d}t}$ into their expression for m. Need not be simplified.
	$m = e^{-t/20}\Big[(-A - 2B)\cos\frac{t}{10}$ $+ (2A - B)\sin\frac{t}{10}\Big]$	**A1**	oe, using the same constants as s.
		[3]	
8(d)	$s(0) = A = 20$ $m(0) = -A - 2B = 20 \Rightarrow B = \ldots$	**M1**	Substitutes $t = 0$, $s = 20$, $m = 20$ into their s and m and solves for both constants.
	$A = 20,\ B = -20$	**A1**	
	$20e^{-t/20}\left(\cos\frac{t}{10} - \sin\frac{t}{10}\right) = 0$ $\Rightarrow \tan\frac{t}{10} = 1 \Rightarrow t = \ldots$	**M1**	Sets their $s = 0$ and solves for t, e.g. via $\tan\frac{t}{10} = k$ or a harmonic form.

Question 8 continued

Question	Scheme	Marks	Notes
	$t = \frac{5\pi}{2} \approx 7.85$; 2028	**A1**	$\frac{5\pi}{2}$ or awrt 7.85, the first positive root, and 2028.
	$m = 20e^{-\pi/8}\left(\cos\frac{\pi}{4} + 3\sin\frac{\pi}{4}\right)$	**M1**	Substitutes their A, B and their t into their m.
	$m = 40\sqrt{2}\,e^{-\pi/8} \approx 38.2$	**A1**	awrt 38.2, or 38, or exact equivalent.
	$s < 0$ for $t > \frac{5\pi}{2}$: impossible No saplings and 38 trees, fewer than 40 $\Rightarrow$ unsuitable in the long term	**E1**	Concludes the model is unsuitable, at least long term, citing their t or m: e.g. $s < 0$ afterwards, or no saplings with fewer than the original 40 trees.
		[7]	
Total for Question 8		**17**	

Question	Scheme	Marks	Notes
9(a)	$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{r}{500}y - \frac{r}{300}x$ $\Rightarrow \frac{r}{300} = \frac{1}{20}$ or $\frac{r}{500} = \frac{3}{100}$	**M1**	Forms the salt balance for A with inflow $\frac{ry}{500}$ and an outflow term in $\frac{x}{300}$, and equates a coefficient with the given equation.
	$r = 15$	**A1**	
		[2]	
9(b)	Inflow to $B = 15 - 3 = 12$	**B1**	
	$\frac{\mathrm{d}y}{\mathrm{d}t} = 12\mu + \frac{3x}{300} - \frac{15y}{500}$ $\Rightarrow 12\mu = 24$	**M1**	Forms the salt balance for B: μ times their inflow rate, $\frac{3x}{300}$ and an outflow term in $\frac{y}{500}$; equates the constant term with 24.
	$\mu = 2$	**A1 AG**	Correct balance with rates 12 and 15, compared with the given equation, leading without error to $\mu = 2$.
		[3]	
9(c)	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t}$ $+ \frac{3}{100}\left(\frac{x}{100} - \frac{3y}{100} + 24\right)$ with $3y = 100\frac{\mathrm{d}x}{\mathrm{d}t} + 5x$	**M1**	Differentiates one equation, substitutes the other and eliminates y (or x) to reach a second-order equation in one variable. Condone coefficient slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{2}{25}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{2500}x = \frac{18}{25}$	**A1**	oe, such as $2500\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 200\frac{\mathrm{d}x}{\mathrm{d}t} + 3x = 1800$.
	$\lambda^2 + \frac{2}{25}\lambda + \frac{3}{2500} = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation from their second-order equation and solves it by factorising, formula, completing the square or calculator.
	$\lambda = -\frac{1}{50},\ -\frac{3}{50}$ $(x =)\ Ae^{-t/50} + Be^{-3t/50}$	**A1**	
	$\frac{3}{2500}k = \frac{18}{25} \Rightarrow k = 600$	**B1**	
	$x = 600 + Ae^{-t/50} + Be^{-3t/50}$	**A1ft**	ft their complementary function plus their constant particular integral, with two arbitrary constants.
	$t = 0$: $\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{1}{20}(300) + \frac{3}{100}(1200) = 21$	**B1**	
	$600 + A + B = 300$ $-\frac{1}{50}A - \frac{3}{50}B =$ "21" $\Rightarrow A = \ldots,\ B = \ldots$	**M1**	Uses $x = 300$ and their $\frac{\mathrm{d}x}{\mathrm{d}t}$ (or $y = 1200$) at $t = 0$ in their general solution and its derivative, and solves for both constants. Condone slips.

Question 9 continued

Question	Scheme	Marks	Notes
	$x = 600 + 75e^{-t/50} - 375e^{-3t/50}$	**A1**	
	$3y = 100\frac{\mathrm{d}x}{\mathrm{d}t} + 5x$ $= 100\left(-\frac{3}{2}e^{-t/50} + \frac{45}{2}e^{-3t/50}\right) + 5x$	**M1**	Substitutes their x and its derivative into the first equation rearranged for y. Condone slips in differentiating exponentials.
	$y = 1000 + 75e^{-t/50} + 125e^{-3t/50}$	**A1**	oe
		[11]	
	Total for Question 9	**16**	

Question	Scheme	Marks	Notes
10(a)	If $B > A$, water is pumped into tank A, so $k(B-A) > 0$ $\Rightarrow k > 0$	**E1**	Links the direction of pumping to the sign of $k(B-A)$ to conclude $k > 0$. One case, $B > A$ or $A > B$, suffices.
		[1]	
10(b)	$\frac{\mathrm{d}^2A}{\mathrm{d}t^2} = \frac{\mathrm{d}B}{\mathrm{d}t} - \frac{\mathrm{d}A}{\mathrm{d}t} + 8t$ $= A - B - \frac{\mathrm{d}A}{\mathrm{d}t} + 8t$	**M1**	Differentiates the first equation with respect to t and substitutes for $\frac{\mathrm{d}B}{\mathrm{d}t}$. Condone a slip in differentiating $4t^2$.
	$A - B = 4t^2 - \frac{\mathrm{d}A}{\mathrm{d}t}$ $\Rightarrow \frac{\mathrm{d}^2A}{\mathrm{d}t^2} = \left(4t^2 - \frac{\mathrm{d}A}{\mathrm{d}t}\right) - \frac{\mathrm{d}A}{\mathrm{d}t} + 8t$	**dM1**	Uses the first equation to eliminate B, leaving only A, its derivatives and t. Condone sign slips.
	$\frac{\mathrm{d}^2A}{\mathrm{d}t^2} + 2\frac{\mathrm{d}A}{\mathrm{d}t} = 4t^2 + 8t$	**A1 AG**	Correct substituted equations leading without error to the printed equation; collected steps accepted.
		[3]	
10(c)(i)	$\lambda^2 + 2\lambda = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any means.
	$(A =)\ C_1 + C_2e^{-2t}$	**A1**	oe, any two arbitrary constants.
		[2]	
10(c)(ii)	$\frac{\mathrm{d}^2A}{\mathrm{d}t^2} + 2\frac{\mathrm{d}A}{\mathrm{d}t} = 4at + 2a + 2b$: no t^2 term (the constant c is in the complementary function)	**E1**	Shows or states that the trial gives no t^2 term, or that its constant term is in the complementary function. Either reason suffices.
		[1]	
10(c)(iii)	$(6at + 2b) + 2(3at^2 + 2bt + c) = 4t^2 + 8t$ $\Rightarrow 6a = 4,\ 6a + 4b = 8,\ 2b + 2c = 0$	**M1**	Differentiates the trial twice, substitutes and equates coefficients of at least two powers of t. Condone slips in differentiation.
	$a = \frac{2}{3},\ b = 1,\ c = -1$	**A1**	
	$A = C_1 + C_2e^{-2t} + \frac{2}{3}t^3 + t^2 - t$	**A1ft**	ft their complementary function and their a, b, c; needs $A =$ and two arbitrary constants.
		[3]	

Question 10 continued

Question	Scheme	Marks	Notes
10(d)	$\frac{\mathrm{d}A}{\mathrm{d}t} = -2C_2e^{-2t} + 2t^2 + 2t - 1$ $t = 0: \ -2C_2 - 1 = 13$	**M1**	Differentiates their general solution, of the form $C_1 + C_2e^{-2t}+$ polynomial, and substitutes $t = 0$, $\frac{\mathrm{d}A}{\mathrm{d}t} = 13$. Condone differentiation slips.
	$t = 0: \ C_1 + C_2 = 52$	**M1**	Substitutes $t = 0$, $A = 52$ into their general solution.
	$C_2 = -7, \ C_1 = 59$	**A1**	
	$A = "59 - 7e^{-1}" + \frac{2}{3}(0.5)^3 + 0.5^2 - 0.5$	**M1**	Substitutes $t = 0.5$, not $t = 30$, into their solution with their constants.
	$A \approx 56.3$ (m^3)	**A1**	awrt 56.3, or $59 - 7e^{-1} - \frac{1}{6}$ oe.
		[5]	
10(e)	$4t^2$, and so A, grows without bound, but the tanks have finite capacity	**E1**	Refers to the unbounded inflow or volume and a physical limit such as tank capacity. Not for 'the model is a simplification' alone.
		[1]	
	Total for Question 10	**16**	

Question	Scheme	Marks	Notes
11(a)	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -6\frac{\mathrm{d}x}{\mathrm{d}t} - 4\frac{\mathrm{d}y}{\mathrm{d}t}$ $= -6\frac{\mathrm{d}x}{\mathrm{d}t} - 4(x-y)$	**M1**	Differentiates the first equation with respect to t and substitutes $\frac{\mathrm{d}y}{\mathrm{d}t} = x - y$. Condone sign slips.
	$4y = 40 - 6x - \frac{\mathrm{d}x}{\mathrm{d}t}$ $\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -6\frac{\mathrm{d}x}{\mathrm{d}t} - 4x + 40 - 6x - \frac{\mathrm{d}x}{\mathrm{d}t}$	**dM1**	Eliminates y using the first equation, giving an equation in x and its derivatives only. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 7\frac{\mathrm{d}x}{\mathrm{d}t} + 10x = 40$	**A1 AG**	Correct substituted equation leading without error to the printed result; combined steps accepted.
		[3]	
11(b)	$\lambda^2 + 7\lambda + 10 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it: factorising with product of constants 10, the formula, or completing the square. Calculator allowed.
	$\lambda = -2, -5$ $(x =)\ Ae^{-2t} + Be^{-5t}$	**A1**	
	Particular integral $x = 4$	**B1**	
	$x = Ae^{-2t} + Be^{-5t} + 4$	**A1ft**	ft their complementary function of the form $Ae^{\alpha t} + Be^{\beta t}$ + their constant particular integral.
	$t = 0:\ \frac{\mathrm{d}x}{\mathrm{d}t} = 40 - 6(1) - 4(1) = 30$ $A + B + 4 = 1, \quad -2A - 5B = 30$	**M1**	Uses x at $t = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}t}$ from the first equation in their general solution and its derivative, forming two equations in A, B. Condone arithmetic slips.
	$\Rightarrow A = \ldots,\ B = \ldots$	**dM1**	Solves their two simultaneous equations for A and B; calculator allowed.
	$x = 5e^{-2t} - 8e^{-5t} + 4$	**A1**	
		[7]	
11(c)	$\frac{\mathrm{d}x}{\mathrm{d}t} = -10e^{-2t} + 40e^{-5t} = 0$ $\Rightarrow e^{3t} = 4 \Rightarrow t = \ldots$	**M1**	Differentiates their x, sets $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and uses logarithms to reach $t = \ldots$. No verification of a maximum needed.
	$t = \frac{1}{3}\ln 4$	**A1**	Or any exact equivalent, such as $\frac{2}{3}\ln 2$.
		[2]	

Question 11 continued

Question	Scheme	Marks	Notes
11(d)	$x = 3 \times 4^{-\frac{2}{3}} + 4$ $40 - 6x - 4y = 0 \Rightarrow y = \ldots$	**M1**	Substitutes their t from (c) into their x and uses $40 - 6x - 4y = 0$ to find y.
	$y = 4 - \frac{9}{2} \times 4^{-\frac{2}{3}}$ $[\approx 2.214]$ 221 ladybirds	**A1**	Awrt 221 ladybirds; $y \approx 2.21$ alone scores A0.
		[2]	
11(e)	For small x, $\frac{\mathrm{d}x}{\mathrm{d}t} \approx 40 - 4y$: eating rate $4y$ does not depend on x, so x can become negative	**E1**	States that the $-4y$ term does not depend on x, so $\frac{\mathrm{d}x}{\mathrm{d}t}$ can stay negative as $x \to 0$: x negative, or aphids eaten when none remain.
		[1]	
	Total for Question 11	**15**	

Question	Scheme	Marks	Notes
12(a)(i)	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = p\frac{\mathrm{d}x}{\mathrm{d}t} - a\frac{\mathrm{d}y}{\mathrm{d}t}$ $= p\frac{\mathrm{d}x}{\mathrm{d}t} - a(qy - bx)$	**M1**	Differentiates the first equation with respect to t and substitutes the second for $\frac{\mathrm{d}y}{\mathrm{d}t}$. Condone sign slips.
	$ay = px - \frac{\mathrm{d}x}{\mathrm{d}t} \Rightarrow$ $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = p\frac{\mathrm{d}x}{\mathrm{d}t} - q\left(px - \frac{\mathrm{d}x}{\mathrm{d}t}\right) + abx$	**dM1**	Uses the first equation to replace y, giving an equation in x and its derivatives only. Condone sign slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - (p+q)\frac{\mathrm{d}x}{\mathrm{d}t} + (pq - ab)x = 0$	**A1**	Correct substituted equation leading without error to the printed equation.
	$x = e^{\lambda t} \Rightarrow \left(\lambda^2 - (p+q)\lambda + (pq-ab)\right)e^{\lambda t} = 0$ $e^{\lambda t} \neq 0 \Rightarrow \lambda^2 - (p+q)\lambda + (pq-ab) = 0$	**M1**	Substitutes $x = e^{\lambda t}$ into the derived equation and cancels $e^{\lambda t} \neq 0$. Stating the auxiliary equation alone scores M0.
	Distinct roots: $e^{\lambda_1 t}$, $e^{\lambda_2 t}$ both solutions $\Rightarrow x = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$	**A1 AG**	Concludes from two distinct roots that $e^{\lambda_1 t}$ and $e^{\lambda_2 t}$ are both solutions, combined with two arbitrary constants. The printed solution alone scores A0.
		[5]	
12(a)(ii)	$\frac{\mathrm{d}x}{\mathrm{d}t} = \lambda_1 Ae^{\lambda_1 t} + \lambda_2 Be^{\lambda_2 t}$ $y = \frac{1}{a}\big[p(Ae^{\lambda_1 t} + Be^{\lambda_2 t})$ $- (\lambda_1 Ae^{\lambda_1 t} + \lambda_2 Be^{\lambda_2 t})\big]$	**M1**	Differentiates x and substitutes x and $\frac{\mathrm{d}x}{\mathrm{d}t}$ into $y = \frac{1}{a}\left(px - \frac{\mathrm{d}x}{\mathrm{d}t}\right)$ oe. Condone sign slips.
	$y = \frac{p-\lambda_1}{a}Ae^{\lambda_1 t} + \frac{p-\lambda_2}{a}Be^{\lambda_2 t}$	**A1 AG**	Correct substituted expression leading without error to the printed form.
		[2]	
12(b)(i)	$\lambda^2 - 0.04\lambda - 0.0005 = 0$ $\Rightarrow \lambda = 0.05,\ -0.01$	**B1**	
	$y = -\frac{1}{2}Ae^{0.05t} + \frac{3}{2}Be^{-0.01t}$ $A + B = x_0,\ -\frac{1}{2}A + \frac{3}{2}B = y_0$ $\Rightarrow A = \ldots,\ B = \ldots$	**M1**	Uses $t = 0$ in x and in their y of the form $kAe^{0.05t} + mBe^{-0.01t}$ from (a)(ii), and solves the pair for A and B.
	$A = \frac{1}{4}(3x_0 - 2y_0),\ B = \frac{1}{4}(x_0 + 2y_0)$ $x = \frac{1}{4}(3x_0 - 2y_0)e^{0.05t}$ $+ \frac{1}{4}(x_0 + 2y_0)e^{-0.01t}$	**A1 AG**	Correct coefficients $-\frac{1}{2}$, $\frac{3}{2}$ in y and the pair solved without error to the printed result.
		[3]	

Question 12 continued

Question	Scheme	Marks	Notes
12(b)(ii)	$y = -\frac{1}{2} \cdot \frac{1}{4}(3x_0 - 2y_0)e^{0.05t}$ $+ \frac{3}{2} \cdot \frac{1}{4}(x_0 + 2y_0)e^{-0.01t}$ $= -\frac{1}{8}(3x_0 - 2y_0)e^{0.05t}$ $+ \frac{3}{8}(x_0 + 2y_0)e^{-0.01t}$	**B1**	Substitutes A and B from (b)(i) into $y = -\frac{1}{2}Ae^{0.05t} + \frac{3}{2}Be^{-0.01t}$ and reaches the printed result without error.
		[1]	
12(c)(i)	$x(25) = 50e^{1.25} + 350e^{-0.25}$ $y(25) = -25e^{1.25} + 525e^{-0.25}$	**M1**	Substitutes $x_0 = 400$, $y_0 = 500$ and $t = 25$ into both results of (b).
	$x \approx 447$, $y \approx 322$	**A1**	awrt 447 and 322.
		[2]	
12(c)(ii)	$50e^{0.05t} + 350e^{-0.01t} = -25e^{0.05t} + 525e^{-0.01t}$	**M1**	Equates their x and y, each of the form $Pe^{0.05t} + Qe^{-0.01t}$ with x_0, y_0 substituted.
	$e^{0.06t} = \frac{7}{3}$	**A1**	oe, such as $75e^{0.06t} = 175$.
	$0.06t = \ln \frac{7}{3} \Rightarrow t = \ldots$	**dM1**	Takes logarithms of their equation of the form $e^{kt} = c$, $c > 0$, to find t.
	$t \approx 14.1$ years	**A1**	awrt 14.1.
		[4]	
12(c)(iii)	$x = 50e^{0.05t} + 350e^{-0.01t} > 0$ for all t so X never disappears	**E1**	States both terms of x are positive for all t and concludes X never disappears.
	$-25e^{0.05t} + 525e^{-0.01t} = 0$ $\Rightarrow e^{0.06t} = 21 \Rightarrow t = \ldots$	**M1**	Sets their $y = 0$ and solves for t.
	$t \approx 50.7$: Y disappears after about 50.7 years	**A1**	Concludes Y disappears, at awrt 50.7 years.
		[3]	
12(d)(i)	$3x_0 = 2y_0$	**B1**	oe, such as $y_0 = \frac{3}{2}x_0$.
		[1]	
12(d)(ii)	$x = \frac{1}{4}(x_0 + 2y_0)e^{-0.01t} = x_0e^{-0.01t}$ $y = \frac{3}{8}(x_0 + 2y_0)e^{-0.01t} = \frac{3}{2}x_0e^{-0.01t}$	**M1**	Uses their relation from (d)(i) to remove the $e^{0.05t}$ term from both x and y.

Question 12 continued

Question	Scheme	Marks	Notes
	Both decrease exponentially to 0, with $y : x = 3 : 2$	**A1**	Both populations decrease to zero, with constant ratio $y : x = 3 : 2$ oe.
		[2]	
	Total for Question 12	**23**	